

I-Logical Approaches Constraint Satisfaction Problems

Foundations of Neuro-Symbolic AI

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Constraint Satisfaction Problems

We now understand tensor networks $\tau^{\mathcal{G}}$ on hypergraphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ as constraint networks:

- ▶ each node $v \in \mathcal{V}$ decorated by a variable X_v with leg dimension m_v
- ▶ each edge $e \in \mathcal{E}$ decorated by a constraint tensor $\tau^e [X_e]$ with leg dimensions m_e

Given an assignment x_e to the variables of a constraint tensor, we say that the **constraint is satisfied**, if $\tau^e [X_e = x_e] \neq 0$.

Constraint Satisfaction Problem (CSP)

Given a constraint network $\tau^{\mathcal{G}}$ decide whether there is an assignment x_v such that all constraints are satisfied, that is $\tau^e [X_e = x_e] \neq 0$ for all edges.

Example: Satisfiability of a knowledge base

- ▶ Atomic formula correspond with variables X_v of dimension $m_v = 2$
- ▶ Formulas are the constraint tensors

Toy accounting example as a CSP

The toy accounting example consists of two constraint tensors:

- ▶ "Exactly one of **Account 1** and **Account 2** is booked.", enforced by the constraint tensor

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} [X_{A1}, X_{A2}]$$

- ▶ "If the **Feature** is present, then **Account 1** is booked.", enforced by the constraint tensor

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_F, X_{A1}]$$

The toy accounting example is satisfiable, since the assignments satisfy all constraints:

- ▶ $(X_{A1}, X_{A2}, X_F) = (0, 1, 0)$
- ▶ $(X_{A1}, X_{A2}, X_F) = (1, 0, 0)$
- ▶ $(X_{A1}, X_{A2}, X_F) = (1, 0, 0)$

Contraction Criterion

Recall: The contraction $\langle f \rangle_{[\emptyset]}$ counts the number of satisfying assignments of the formula f .

More general:

- ▶ The contraction $\langle \tau^e [X_e] \rangle_{[\emptyset]}$ counts the number of assignments to the variables X_e satisfying the constraint.
- ▶ The contraction $\langle \{ \tau^e [X_e] : e \in \mathcal{E} \} \rangle_{[\emptyset]}$ counts the number of assignments to all variables satisfying all constraints.

Contraction criterion

A CSP is satisfiable, if and only if its contraction does not vanish, that is

$$\langle \{ \tau^e [X_e] : e \in \mathcal{E} \} \rangle_{[\emptyset]} \neq 0.$$

While global contraction is a generic solver, it cannot be an efficient one.

Outline

- ▶ Constraint Satisfaction Problems
- ▶ **Sudoku**
- ▶ Generalized unit propagation
- ▶ Graph coloring
- ▶ Backtracking search

Example: Sudoku

Consider grids of $n^2 \times n^2$ cells assigned by numbers in $[n^2]$, where usually $n = 3$ but for visualization purposes we will take $n = 2$.

Constraints:

- ▶ Each cell contains exactly one digit
- ▶ Each digit appears exactly once in each row, column and square

Example of a $n = 2$ Sudoku with initial clues and unique solution:

0			
	1		
		3	
	2		

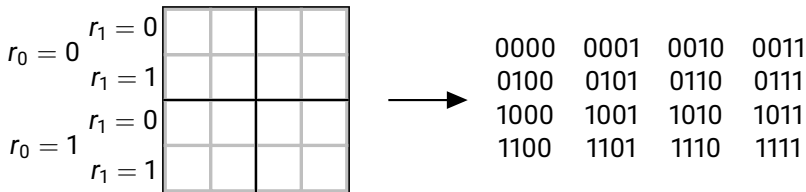
=

0	3	2	1
2	1	0	3
1	0	3	2
3	2	1	0

Representation of Sudoku

We enumerate each cell by tuples $r_0r_1c_0c_1 \in [n]^4$, where (r_0, r_1) select the $r_0 \cdot n + r_1$ th row and (c_0, c_1) the $c_0 \cdot n + c_1$ th column.

Example of a $n = 2$ grid:



For each possible assignment i to each cell $r_0r_1c_0c_1$ we define a hidden variable $X_{r_0r_1c_0c_1i}$, which is 1 if the cell $r_0r_1c_0c_1$ is assigned by i and 0 else.

In total the set of atomic variables is:

$$\{X_{r_0r_1c_0c_1i} : r_0, r_1, c_0, c_1 \in [n], i \in [n^2]\}$$

Sudoku constraint tensors

Sudoku constraints enforce the "exactly-one" condition by tensors $\tau [X_{\mathcal{U}}]$ with indices

$$\tau [X_{\mathcal{U}} = x_{\mathcal{U}}] = \begin{cases} 1 & \text{if } \sum_{u \in \mathcal{U}} x_u = 1 \\ 0 & \text{else} \end{cases}$$

We choose $\mathcal{U}$ differently for each constraint:

- ▶ **Position constraints:** $\mathcal{U} = \{(r_0 r_1 c_0 c_1 i) : i \in [n^2]\}$ for each cell $(r_0 r_1 c_0 c_1)$
- ▶ **Row constraints:** $\mathcal{U} = \{(r_0 r_1 c_0 c_1 i) : c_0, c_1 \in [n]\}$ for each row $(r_0 r_1)$ and each digit i
- ▶ **Column constraints:** $\mathcal{U} = \{(r_0 r_1 c_0 c_1 i) : r_0, r_1 \in [n]\}$ for each column $(c_0 c_1)$ and each digit i
- ▶ **Square constraints:** $\mathcal{U} = \{(r_0 r_1 c_0 c_1 i) : r_1, c_1 \in [n]\}$ for each square $(r_0 c_0)$ and each digit i

Decomposition of Sudoku constraints

Idea: Introduce for each constraint a decomposition variable $I \in [n^2]$ indicating which of the n^2 atomic variables X in the constraint gets the 1 assignment.

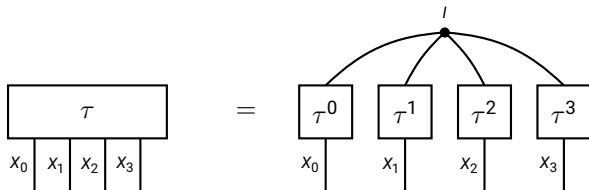
This is enforced for each $i \in [n^2]$ by constraint matrices

$$\tau^i [X_i, I] = \epsilon_1 [X] \otimes \epsilon_i [I] + \epsilon_0 [X] \otimes (\mathbb{I} [I] - \epsilon_i [I]) .$$

For example:

$$\tau^2 [X, I] = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} [X, I]$$

In total we have a decomposition (in the so-called CP-format):



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- ▶ Sudoku
- ▶ **Generalized unit propagation**
- ▶ Graph coloring
- ▶ Backtracking search

Disentangling the constraint network

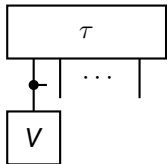
Idea: When a CSP has a unique solution x_v , then the contracted tensor is elementary:

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_{[X_v]} = \bigotimes_{v \in \mathcal{V}} \epsilon_{x_v} [X_v]$$

Aim: Manipulate the network to **disentangle** the constraints.

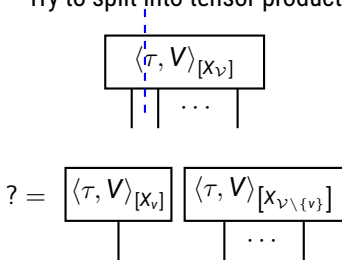
1.Step:

Contract vector on constraint

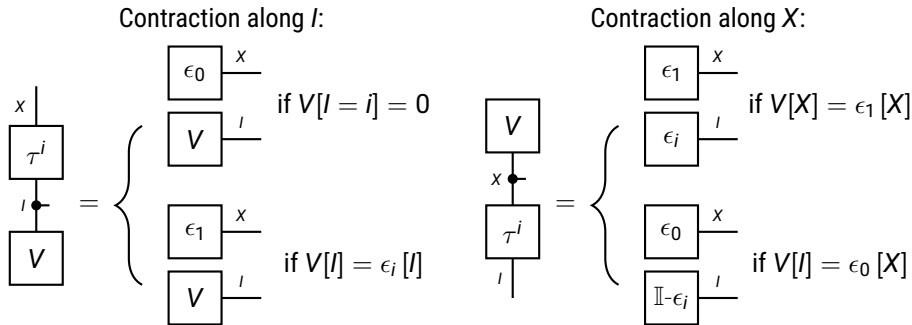


2.Step:

Try to split into tensor product



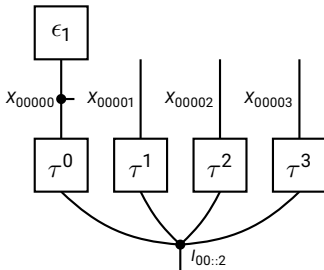
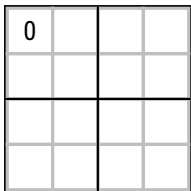
Disentangling properties of Sudoku constraint matrices



**These disentangling matrix-vector contractions are exploited to solve
(simple) Sudoku puzzles.**

Disentangling a known position in Sudoku

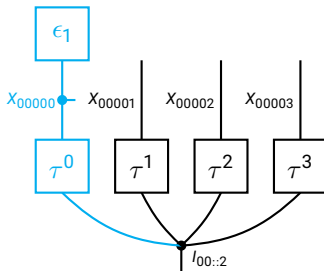
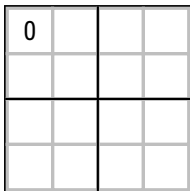
Example: When 0 is assigned at position 0000, then no further digit can be assigned to that position.



We disentangle each atomic variable corresponding with a known position.

Disentangling a known position in Sudoku

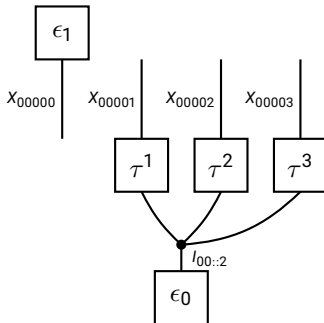
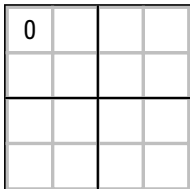
Example: When 0 is assigned at position 0000, then no further digit can be assigned to that position.



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Disentangling a known position in Sudoku

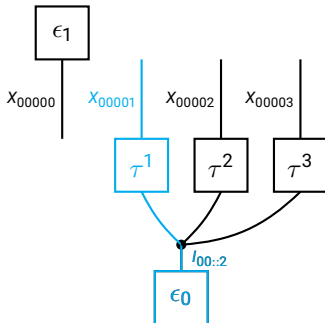
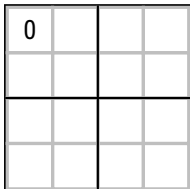
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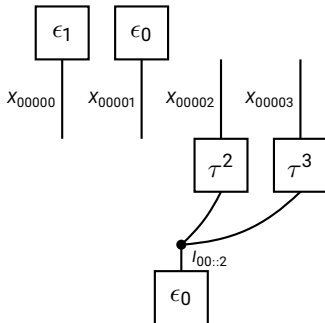
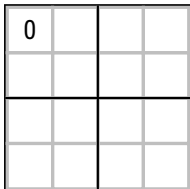
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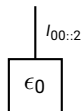
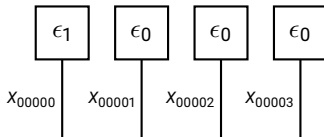
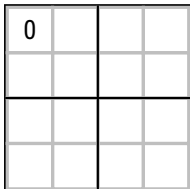
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We disentangle each atomic variable corresponding with a known position.

Disentangling a known position in Sudoku

Example: When 0 is assigned at position 0000, then no further digit can be assigned to that position.



We disentangle each atomic variable corresponding with a known position.

Generalized unit propagation

Until convergence, we

- ▶ Contract a vector to each neighboring tensor.
- ▶ Try to split the resulting tensor into a vector and a tensor.
- ▶ If success and the remaining tensor is a vector: Repeat for the remaining vector.

Comparison with unit propagation so far:

- ▶ Works with arbitrary constraints (shape, values), while so far on unit clauses
- ▶ Performs sub-contraction and tries to split the resulting tensor

Generalized unit propagation algorithm

Require: Constraint network $\tau^{\mathcal{G}}$

Ensure: Equivalent constraint network $\tau^{\mathcal{G}}$, with potentially less connectivity in $\mathcal{G}$

Initialize a queue Q of unary edges in $\mathcal{G}$

while Q not empty **do**

$\{v\} = Q.pop()$

for $e \in \mathcal{E} : v \in e$ **do**

$\tau^e[X_e] \leftarrow \langle \tau^e[X_e], \tau^{\{v\}}[X_v] \rangle_{[X_e]}$

 Try to split $\tau^e[X_e]$ into a vector v and a tensor $e \setminus \{v\}$

if Split successful and $|e \setminus \{v\}| = 1$ **then**

$Q.push(e \setminus \{v\})$

end if

end for

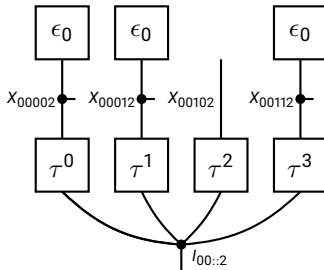
end while

return $\tau^{\mathcal{G}}$

Matrix-vector example

Find the 2 in the first row:

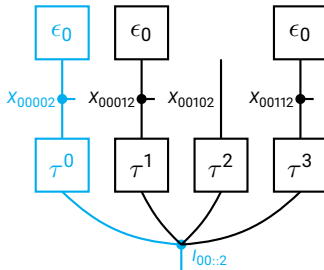
0	3		1



Matrix-vector example

Find the 2 in the first row:

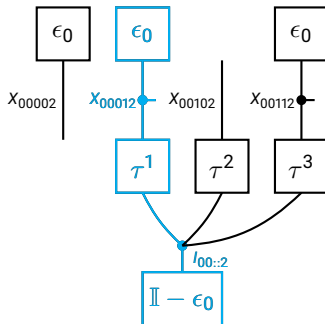
0	3		1



Matrix-vector example

Find the 2 in the first row:

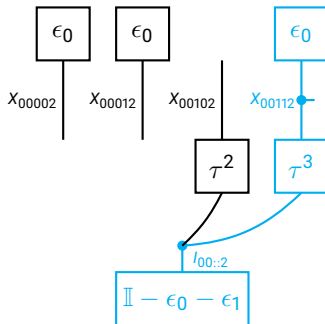
0	3		1



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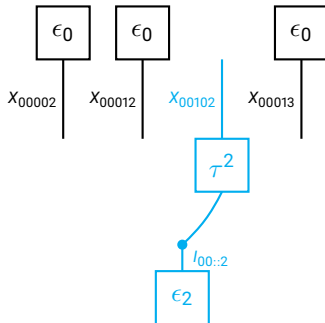
0	3		1



Matrix-vector example

Find the 2 in the first row:

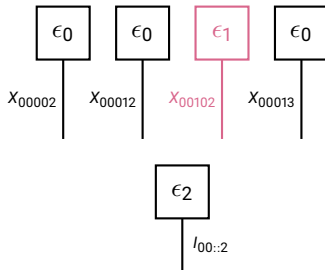
0	3		1



Matrix-vector example

Find the 2 in the first row:

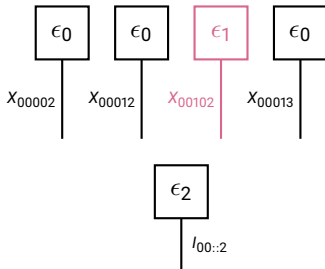
0	3	2	1



Matrix-vector example

Find the 2 in the first row:

0	3		1



With these matrix-vector contraction schemes, the example Sudoku can be solved:

0			
	1		
		3	
	2		

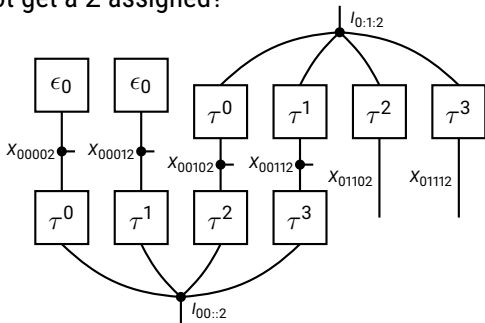
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0	3	2	1
2	1	0	3
1	0	3	2
3	2	1	0

Limitations

Example: Which cells cannot get a 2 assigned?

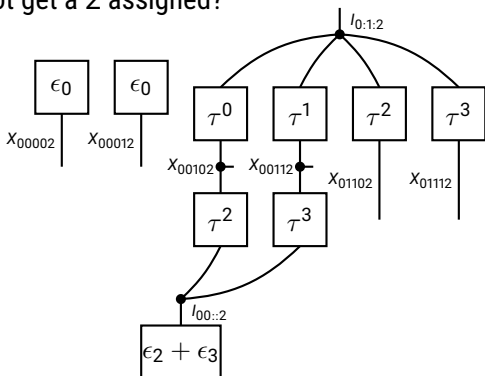
0	3		



Limitations

Example: Which cells cannot get a 2 assigned?

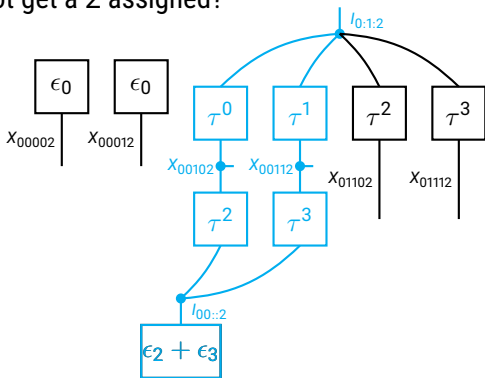
0	3		



Limitations

Example: Which cells cannot get a 2 assigned?

0	3		



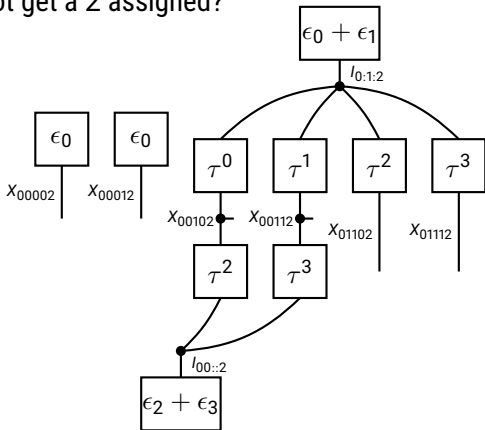
We need a contraction of four constraint matrices, to continue!

On the Nicolli100 benchmark: Only 49% solved by generalized unit propagation.

Limitations

Example: Which cells cannot get a 2 assigned?

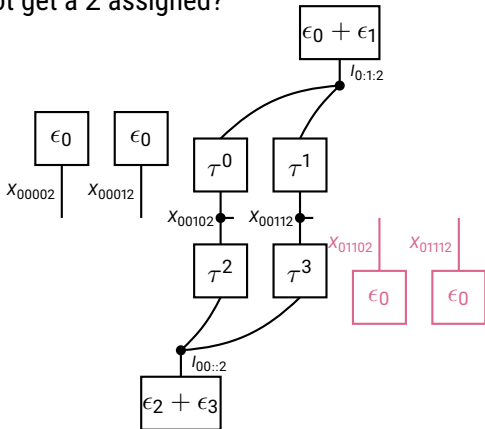
0	3		



Limitations

Example: Which cells cannot get a 2 assigned?

0	3		
		2	2



Result: Two cells can be excluded for 2.

Outline

- ▶ Constraint Satisfaction Problems
- ▶ Sudoku
- ▶ Generalized unit propagation
- ▶ **Graph coloring**
- ▶ Backtracking search

Example: Graph coloring

Graph coloring problem

Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and m colors, find a coloring of the nodes, such that adjacent nodes have different colors.

Example: Paint the states of australia



- ▶ How many colors are required?
- ▶ How many possible colorings can be done?

Representation of the graph coloring problem

We model the problem:

- ▶ Enumerate the colors by the state set $[m] = \{0, \dots, m-1\}$
- ▶ For each node $v \in \mathcal{V}$ we define a variable X_v

To each edge $e = \{v_0, v_1\}$ the "different" constraint matrix

$$\tau^{v_0 \neq v_1} [X_{v_0}, X_{v_1}] = \mathbb{I} [X_{v_0}, X_{v_1}] - \delta [X_{v_0}, X_{v_1}] .$$

with off-diagonal support enforces different color constraint.

$$\tau^{v_0 \neq v_1} [X_{v_0}, X_{v_1}] = \begin{pmatrix} 0 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 0 \end{pmatrix} [X_{v_0}, X_{v_1}]$$

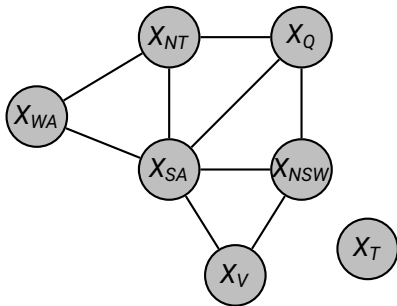
Example: Painting states of australia

Let us paint the states of australia with 3 colors:

- ▶ 7 color variables X_v of leg dimension $m = 3$
- ▶ 9 edges between neighbored states v_0 and v_1 , which are decorated by the 3x3 constraint matrix

For each edge $\{v_0, v_1\} \in \mathcal{E}$ we have a constraint matrix:

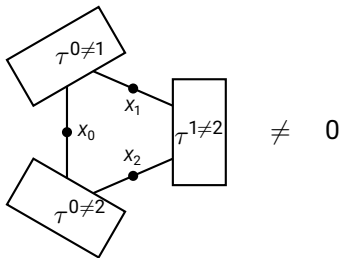
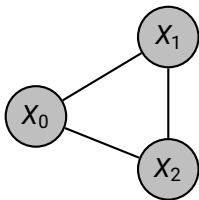
$$\begin{aligned} & \tau^{v_0 \neq v_1} [X_{v_0}, X_{v_1}] \\ &= \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} [X_{v_0}, X_{v_1}] \end{aligned}$$



Example: Cycle of three nodes

Question: Is a cycle of three nodes colorable by two colors?

Consider a cycle of three nodes: When colorable then the following holds:



However: We cannot detect the unsatisfiability by contractions of two matrices.

In general: Cycles with odd numbers need three colors, and of even need two colors.

Example: Cycle of three nodes

Let us allow for three colors (i.e. $m_0 = m_1 = m_2 = 3$). The contraction of the three "different" constraints is non-vanishing, and we can read off the solutions:

$$\langle \tau^{0 \neq 1} [X_0, X_1], \tau^{0 \neq 2} [X_0, X_2], \tau^{1 \neq 2} [X_1, X_2] \rangle_{[X_0, X_1, X_2]} =$$

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} [X_0, X_1, X_2]$$

Interpretation: The contracted tensor has 6 non-vanishing entries, which correspond to the 6 possible colorings of the three nodes by three colors.

Disentangling properties of the "different" constraint

When contracting the "different" constraint with a vector, we get

$$\begin{aligned} \mathbb{I}_{\neq 0} \circ \left\langle V[X_{v_0}], \tau^{v_0 \neq v_1} [X_{v_0}, X_{v_1}] \right\rangle_{[X_{v_1}]} \\ = \begin{cases} \mathbb{I}[X_{v_1}] - \epsilon_y[X_{v_1}] & \text{if } V[X_{v_0}] \text{ only at } y \text{ supported} \\ \mathbb{I}[X_{v_1}] & \text{else} \end{cases} \end{aligned}$$

The "different" constraint has weakest disentangling properties:

Disentangling only possible, when the vector is basis (i.e. the color of one node is known).

Consequence: Unit propagation helps only, when the color of one node v_0 is known:

But where to find the basis vectors necessary for disentangling the color constraints?

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- ▶ Constraint Satisfaction Problems
- ▶ Sudoku
- ▶ Generalized unit propagation
- ▶ Graph coloring
- ▶ **Backtracking search**

Idea of backtracking

Problem: Some constraints do not disentangle. This might be due to the fact, that multiple solutions are available (which do not build product sets).

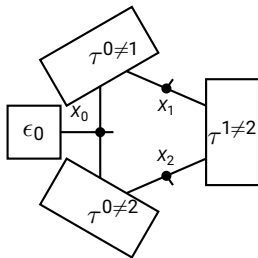
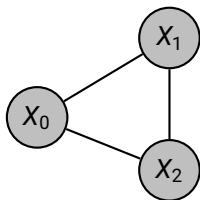
Solution: Backtracking search

- ▶ Guess an assignment to a variable, and add the one-hot encoding $\epsilon_i [X_v]$.
- ▶ Any constraint tensor to an edge e with $v \in e$ disentangles (maybe more by unit propagation).
- ▶ If an inconsistency is detected (i.e. a vanishing constraint tensor), guess an alternative assignment.
- ▶ Iterate until a non-vanishing elementary constraint network is reached, from which (possibly multiple) solutions can be constructed by local solutions.

Example: Coloring a cycle of three nodes

To show that a cycle of three nodes is not colorable by two colors, we can use backtracking:

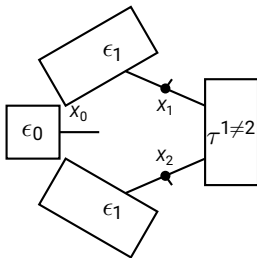
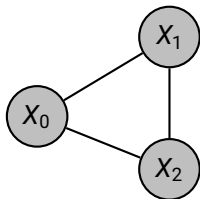
- ▶ Guess X_0 to be 0, and get an inconsistency.
- ▶



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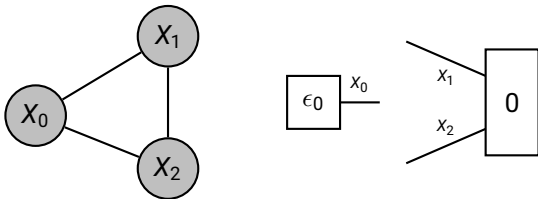
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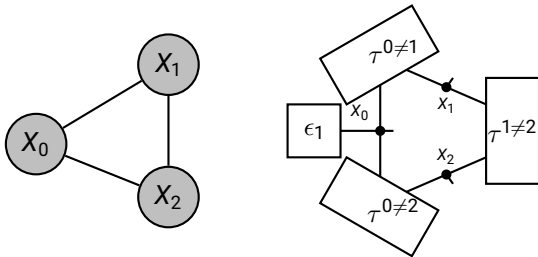
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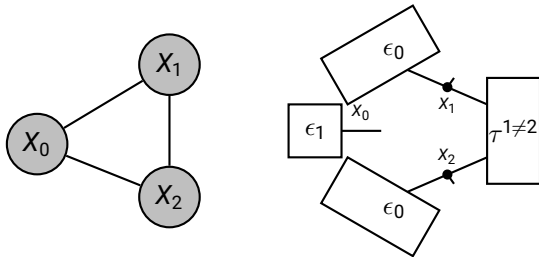
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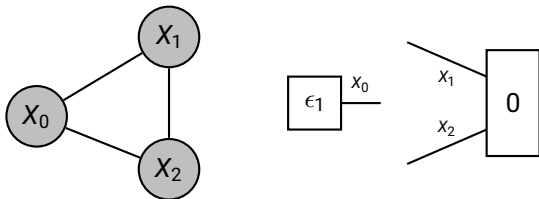
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- ▶ Guess X_0 to be 0, and get an inconsistency.
- ▶ Guess X_0 to be 1, and get an inconsistency.



Backtracking algorithm

Require: Constraint network $\tau^{\mathcal{G}}$

Ensure: One-hot encoding of a solution, or *False* if not satisfiable

Choose a node v in a non-unitary edge e

for $x_v \in [m_v]$ **do**

$\tilde{\tau}^{\mathcal{G}} \leftarrow \tau^{\mathcal{G}} \cup \{\epsilon_{x_v} [X_v]\}$

$\tilde{\tau}^{\mathcal{G}} \leftarrow \text{Simplify}(\tilde{\tau}^{\mathcal{G}})$

(differing realization)

if $\tilde{\tau}^{\mathcal{G}}$ vanishing **then**

return *False*

end if

$\tilde{\tau}^{\mathcal{G}} \leftarrow \text{Backtrack}(\tau^{\mathcal{G}} \cup \{\epsilon_{x_v} [X_v]\})$

if $\tilde{\tau}^{\mathcal{G}}$ not *False* **then**

return $\tilde{\tau}^{\mathcal{G}}$

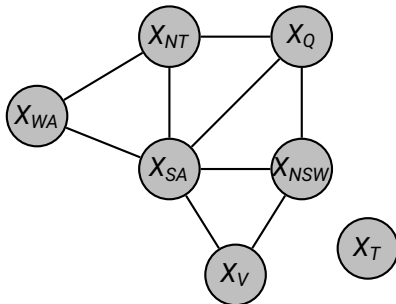
end if

end for

return *False*

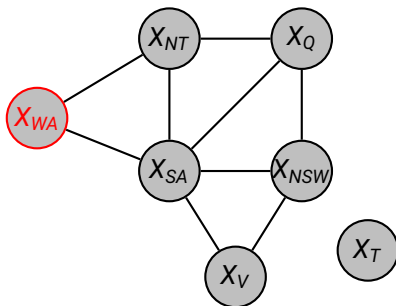
Example: Painting australian states

Take three colors: red, blue, green



Example: Painting australian states

Take three colors: red, blue, green

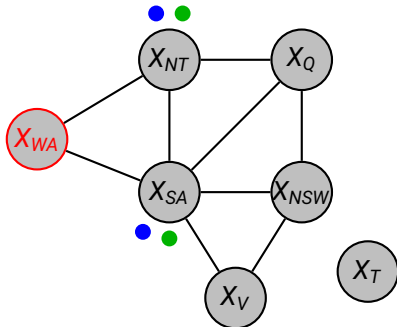


► Guess X_{WA} to be red.



Example: Painting australia's states

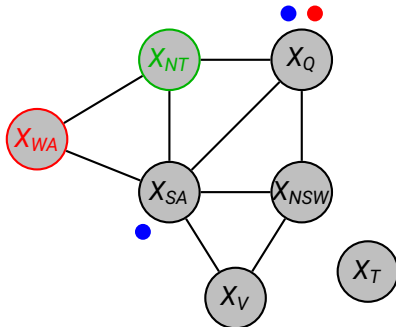
Take three colors: red, blue, green



- ▶ Guess X_{WA} to be red.
- ▶ GUP finds consequences for X_{NT} and X_{SA} : red or green
- ▶
- ▶
- ▶
- ▶

Example: Painting australia's states

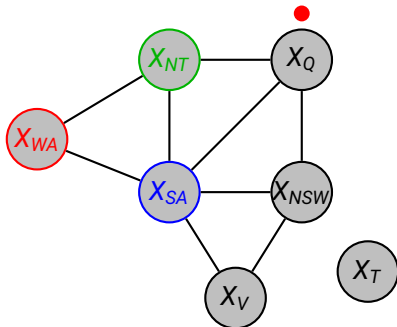
Take three colors: red, blue, green



- ▶ Guess X_{WA} to be red.
- ▶ GUP finds consequences for X_{NT} and X_{SA} : red or green
- ▶ Guess X_{NT} to be green.
- ▶
- ▶
- ▶

Example: Painting australian states

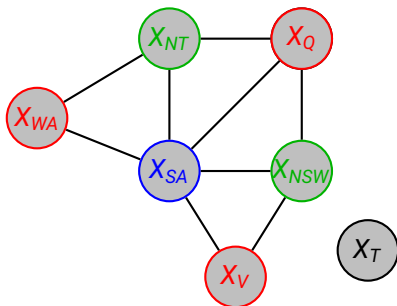
Take three colors: red, blue, green



- ▶ Guess X_{WA} to be red.
- ▶ GUP finds consequences for X_{NT} and X_{SA} : red or green
- ▶ Guess X_{NT} to be green.
- ▶ GUP determines X_{SA} to be blue.
- ▶
- ▶

Example: Painting australian states

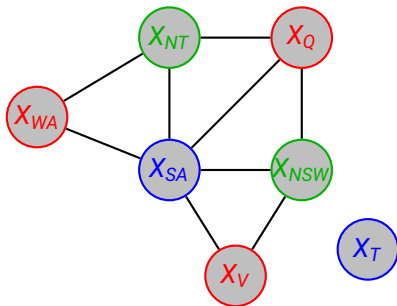
Take three colors: red, blue, green



- ▶ Guess X_{WA} to be red.
- ▶ GUP finds consequences for X_{NT} and X_{SA} : red or green
- ▶ Guess X_{NT} to be green.
- ▶ GUP determines X_{SA} to be blue.
- ▶ GUP determines also X_Q , X_{NSW} , X_V .
- ▶

Example: Painting australian states

Take three colors: red, blue, green



- ▶ Guess X_{WA} to be red.
- ▶ GUP finds consequences for X_{NT} and X_{SA} : red or green
- ▶ Guess X_{NT} to be green.
- ▶ GUP determines X_{SA} to be blue.
- ▶ GUP determines also X_Q , X_{NSW} , X_V .
- ▶ The variable X_T is disconnected and can be chosen arbitrarily.